Muhammad Ariff Bin Abdul Razak

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WORK EXPERIENCE

Machine Learning & Fullstack Engineer (Intern), KLASS Engineering and Solutions
Sep 2025 – Aug 2026

- Reproduced and benchmarked state-of-the-art AI-generated video and audio detectors (NSG-VD, D3, ReStraV, Speech-Forensics) on SLURM-based HPC, achieving 90.03 AUROC on the NSG-VD replication.
- Integrated optical flow (RAFT), segmentation (SAM 2), and point-tracking (CoTracker) algorithms to analyze physical motion derivatives in video sequences for mission-critical data.
- Engineered a data generation pipeline chaining video extension (Wan2.2), voice cloning (Chatterbox), and lip-synchronization (LatentSync) models to curate a balanced 1,200-video benchmark dataset.
- Evaluated audio-visual synchronization as a deepfake detection signal, selecting a ~0.3 operating threshold through ROC analysis.
- Managed GPU-accelerated experimentation on shared SLURM clusters (NVIDIA H200 nodes) via Docker containers, running sanity-check protocols on custom validation splits before committing cluster time to full-scale benchmarking runs.
- Built React frontend components and FastAPI backend services with WebSockets for real-time streaming responses, deploying multi-model AI endpoints for an internal AI chatbot platform.

Student Research Engineer, Singapore Institute of Technology (SIT)
Aug 2024 – May 2025

- Designed and built an experimental temperature chamber simulating human body heat signatures and extreme physical environments for faculty-led embodied AI and robotics hardware testing.
- Engineered a closed-loop, thermistor-based IoT control system to hold chamber temperature at setpoint, enabling high-fidelity data collection.
- Established repeatable testing and data collection protocols for faculty-led embodied AI and robotics hardware projects.

Research Assistant (Full Time / Volunteer), SUTD – ROAR Lab
Aug 2022 – May 2024

- Developed perception software processing 3D LiDAR data for autonomous mobile robots, designing architectures integrating Embedded C++, ROS, and sensor fusion.
- Led the design of the full intelligent lighting control system for Panthera, an autonomous vehicle developed under National Robotics Programme funding, implementing it as a rosserial node on an Arduino that subscribed to cmd_vel, IMU, and vehicle-state topics under an RTOS to drive turn, warning, and hazard signaling; the platform passed T1 autonomous testing at CETRAN.
- Built a predictive safety model to detect a stair-climbing robot's fall risk, deploying Long Short-Term Memory networks (LSTMs) over sensor time-series data.

Trooper & Armourer, Singapore Police Force – Protective Security Command
Jan 2021 – Nov 2022 (Full-Time National Service)

- Performed protective security duties as part of a deployed team, operating under strict standard operating procedures to protect strategic locations and major national events across 22 months of full-time service.
- Served as unit Armourer alongside trooper duties, managing the issuance, return, and accountability of arms and equipment, and maintaining audit-ready records against regulated inventory procedures.

Robotics Engineer Intern, SOLUSTAR
May 2022 – Aug 2022

- Designed and debugged PCBs and circuitry for a disinfectant robot.
- Performed maintenance on 3D printers and developed custom Marlin firmware in C++ to improve print reliability.
- Troubleshot Linux software issues for the VAL2 robot.

Robotics Instructor & Curriculum Developer, VM Education / Playfacto
Jul 2020 – Dec 2020

- Pioneered and designed the new coding curriculum for the MODI (Luxrobo) platform in Singapore, publishing 5 projects of varying difficulty.
- Facilitated training for incoming MODI instructors, documenting solutions to common teaching difficulties.

Technology Instructor (Part Time / Full Time), Nullspace Group
Nov 2016 – Present

- Instruct students aged 5–17 in STEM-focused courses covering computational thinking, programming (C++, Python), robotics, and microcontrollers, as an MOE-certified instructor from age 18.
- Selected to conduct the Computer Enrichment Program (CEP) at GEP schools in Singapore, completing over 1,000 teaching hours.

- Achieved MOE (Ministry of Education) Instructor Certification at age 18 (Valid Oct 2018 – Oct 2026).

PROJECTS

1. Capstone (MET4004) — Acoustic Noise Reduction for Drones — Project Lead — Software & Electronics (KLASS)
Aug 2026 – Apr 2027

- Leading a 3-person team to deliver a proof-of-concept indoor drone for industry sponsor KLASS, targeting a 78 dBA ceiling at a 1 m reference during steady hover, 10-minute hover endurance, and a 370 mm square airframe envelope.
- Owning the software and electronics workstream, selecting a low-noise propulsion stack of 5-inch injection-molded toroidal propellers, custom ducts with acoustic lining and serrated trailing edges, 1300 KV low-RPM motors, and AM32 ESC PWM frequency tuning to shift tonal whine into less perceptible bands.
- Specified the avionics and power architecture (Holybro Pixhawk 6C Mini, PM02 V2 power module, dedicated 150 A power distribution board), yielding a derated 3.3:1 thrust-to-weight ratio at 1,200 g all-up weight for stable station-keeping in GPS-denied indoor flight.
- Designed the acoustic validation campaign to the Von Karman Institute four-microphone methodology with TU Berlin distance normalization to a 1 m reference, enabling direct benchmarking against HKUST's 90.52 dBA commercial quadcopter baseline and VKI's DJI Phantom 3 SE measurements.
- Specified psychoacoustic post-processing in Python (MoSQITo) under both Zwicker (ISO 532-1) and Moore–Glasberg (ISO 532-2) loudness standards, quantifying perceived annoyance beyond raw sound-pressure level.

2. Autonomous Trolley Collection Robot

- Developed an autonomous robot for collecting stray trolleys on an STM32F7 microcontroller running an RTOS, implementing concurrent tasks for obstacle detection and joystick control over a custom serial communication protocol.

3. Custom 3D Printer Development

- Built a custom 3D printer from off-the-shelf modules, developing custom firmware in C++ (Marlin) and tuning it for print reliability and dimensional accuracy.

4. AI-Generated Video & Audio Forensics Benchmarking
Sep 2025 – Aug 2026

- Reproduced and benchmarked state-of-the-art AI-generated video and audio detectors (NSG-VD, D3, ReStraV, Speech-Forensics) on SLURM-based HPC, achieving 90.03 AUROC on the NSG-VD replication.
- Integrated optical flow (RAFT), segmentation (SAM 2), and point-tracking (CoTracker) algorithms to analyze physical motion derivatives in video sequences for mission-critical data.
- Evaluated audio-visual synchronization as a deepfake detection signal, selecting a ~0.3 operating threshold through ROC analysis.

5. Synthetic Multimodal Deepfake Dataset Pipeline
Sep 2025 – Aug 2026

- Engineered a data generation pipeline chaining video extension (Wan2.2), voice cloning (Chatterbox), and lip-synchronization (LatentSync) models to curate a balanced 1,200-video benchmark dataset.

6. Multi-Model AI Endpoint Platform
Sep 2025 – Aug 2026

- Built React frontend components and FastAPI backend services with WebSockets for real-time streaming responses, deploying multi-model AI endpoints for an internal AI chatbot platform.
- Managed GPU-accelerated experimentation on shared SLURM clusters (NVIDIA H200 nodes) via Docker containers, running sanity-check protocols on custom validation splits before committing cluster time to full-scale benchmarking runs.

7. Stair-Climbing Robot Fall-Risk Prediction (LSTM)
Aug 2022 – May 2024

- Built a predictive safety model to detect a stair-climbing robot's fall risk, deploying Long Short-Term Memory networks (LSTMs) over sensor time-series data.

8. Panthera Autonomous Vehicle — Intelligent Lighting System
Aug 2022 – May 2024

- Led the design of the full intelligent lighting control system for Panthera, an autonomous vehicle developed under National Robotics Programme funding, implementing it as a rosserial node on an Arduino that subscribed to cmd_vel, IMU, and vehicle-state topics under an RTOS to drive turn, warning, and hazard signaling; the platform passed T1 autonomous testing at CETRAN.

9. Experimental Thermal Chamber for Embodied AI Testing
Aug 2024 – May 2025

- Designed and built an experimental temperature chamber simulating human body heat signatures and extreme physical environments for faculty-led embodied AI and robotics hardware testing.
- Engineered a closed-loop, thermistor-based IoT control system to hold chamber temperature at setpoint, enabling high-fidelity data collection.

SKILLS / COMPETENCIES

AI, Vision & Audio Analytics: Deep Learning, Computer Vision, Multimodal ML, Deepfake Detection, Generative AI, OpenCV, Optical Flow (RAFT), Segmentation (SAM 2), Point-Tracking (CoTracker)
Machine Learning Frameworks: PyTorch, TensorFlow, Keras, Hugging Face Transformers, CNNs, DNNs, LSTMs, YOLO, Vision Transformers, Reinforcement Learning, Model Benchmarking, Zero-shot Learning
Robotics, UAV & Autonomous Systems: Embodied AI, ROS, SLAM (3D LiDAR), Perception, Sensor Fusion, Visual-Inertial Odometry (VIO), Autonomous Mobile Robots, Quadcopter Design, UAV Flight Testing, UAV Aeroacoustics, Acoustic Noise Reduction, PX4 Autopilot, ArduPilot, Pixhawk
Embedded, Hardware & Test: Embedded C / C++, RTOS, Microcontrollers (STM32, Arduino), PID Control, Closed-Loop IoT Control, PCB Design, Firmware Development, Hardware Prototyping, Experimental Design, Test & Validation
Software & Fullstack: Python (NumPy, Pandas), C++17, C#, Java, TypeScript, React, Zustand, FastAPI, OpenAPI, WebSockets, Git, Linux/Ubuntu, WSL
Infrastructure & Cloud Tech: SLURM, GPU Computing (CUDA), NVIDIA HPC Clusters (H200 nodes), Docker, Dev Containers, ONNX Runtime, uv, IoT Integrations, Agile/Scrum Methodologies

EDUCATION

DigiPen Institute of Technology & SIT
Aug 2023 – May 2027

- B.Eng. Mechatronics Systems
- Relevant Coursework: Mechanics, Control Systems, Robotics, Embedded Systems, Machine Learning, Systems and Software Engineering

Ngee Ann Polytechnic
Sep 2022 – Sep 2023

- Specialist Diploma in Robotics Engineering (SGInnovate PowerX)
- Relevant Coursework: Autonomous Robot Navigation, Machine Learning/Deep Learning for Robotics, Computer Vision

Singapore Polytechnic
Apr 2017 – Sep 2020

- Diploma in Mechatronics and Robotics
- Internship grade of B+; coursework in Microcontroller Applications, PLCs, and Robotics Integration

AWARDS / ACHIEVEMENTS / CERTIFICATES

- MOE (Ministry of Education) Instructor Certification (Achieved at age 18) — Valid Oct 2018 – Oct 2026
- Google Crash Course on Python
- Google The Bits and Bytes of Computer Networking
- Coursera Machine Learning

VOLUNTEERING & COMMUNITY SERVICE

ROSCon 2025 Singapore (Intrinsic / Open Robotics)

- Supported event operations, contributing to attendee experience, workshops, and logistics.

SG Makers Against COVID-19

- Initiated a pandemic response project to 3D-print ear savers for SingHealth and the local community, distributing over 10,000 units (Featured on Channel News Asia).